LED TORCHLIGHT

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Most commercial torchlights use three AA or AAA cells connected in series, which produces 4.5V to drive white LEDs using a current-limiting resistor. These flashlights do not drive the LEDs efficiently due to lack of a current-control circuit. Presented here is a circuit that drives high, bright white LEDs from a single dry cell.

Circuit and working

ZXSC380 is a highly-integrated, surface-mount device (SMD), single- or multi-cell LED driver for applications where step-up voltage conversion from a very low input voltage is required. It requires only an external inductor to generate constant current pulses that are ideal for driving single or multiple LEDs over a wide range of operating voltages. It provides simple-to-use, low-cost, space-saving and easy-to-layout solutions.

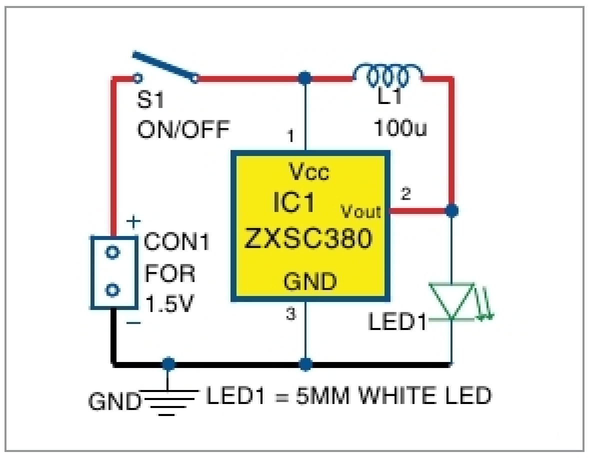

Fig. 1: Torchlight circuit in normal mode

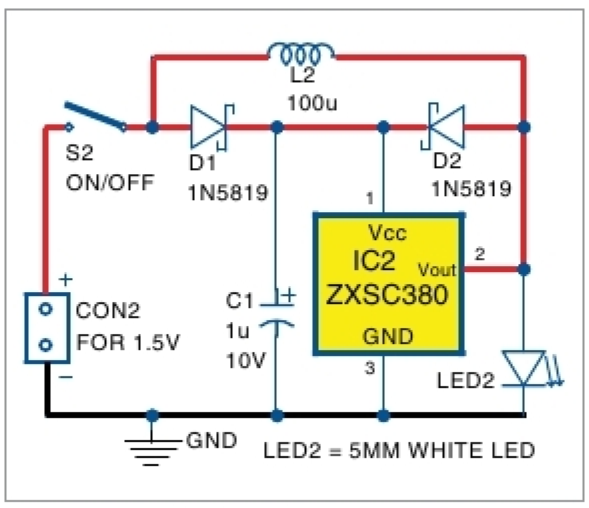

Fig. 2: Torchlight circuit in bootstrap mode

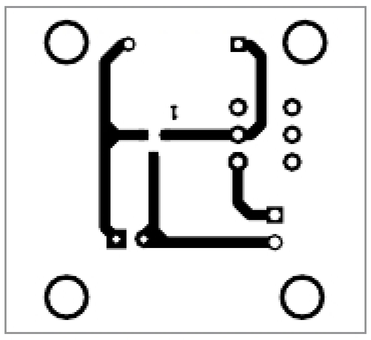

Fig. 3: Actual-size PCB layout of torchlight in normal mode

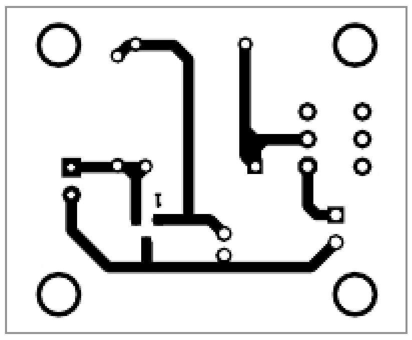

Fig. 5: Actual-size PCB layout of torchlight in bootstrap mode

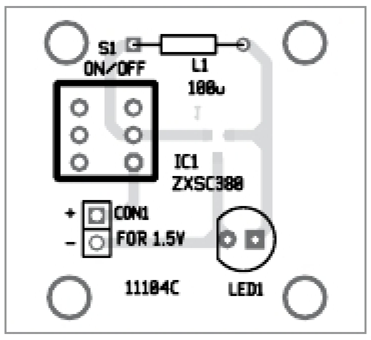

Fig. 4: Components layout for normal mode

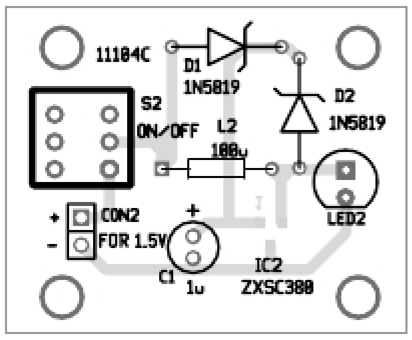

Fig. 6: Components layout for bootstrap mode

PARTS LIST	
Semiconductors:	
IC1, IC2	- ZXSC380 LED driver
D1, D2	- 1N5819 Schottky diode
LED1, LED2	- 5mm LED white LED
Capacitors:	
C1	- 1µF, 10V electrolytic
Miscellaneous:	
CON1, CON2	- 2-pin connector
S1, S2	- On/off switch
L1, L2	- 100µH inductor
	- 1.5V cell (two numbers)

The LED torchlight circuit consists of two modes: normal and bootstrap.

Normal mode. The circuit diagram of the torchlight in normal mode is shown in Fig. 1. In this mode, the circuit requires only one 100µH inductor (L1) to drive LED1. Switch S1 is used to turn on/off IC1. When S1 is closed, IC1 and L1 perform step-up operation and LED1 glows.

Bootstrap mode. The circuit diagram of the torchlight in bootstrap mode is shown in Fig. 2. In this mode, the circuit requires two additional Schottky barrier diodes (D1 and D2) and capacitor C1. When switch S2 is closed, IC2 is powered through diode D1. Inductor L2 charges till the internal switching transistor of IC2 is turned on.

After the internal transistor is turned off, voltage across L2 reverses its polarity and voltage is given back to V_{CC} (pin1) of IC2 through diode D2. Hence, under 1V is required to drive LED2 after startup.

In bootstrap mode, IC2 can be operated down to 0.7V typical after initial successful startup. Operation down to 0.7V typical allows further cell energy to be extracted beyond the typically quoted end of battery voltage of approximately 0.9V. This prolongs battery use time. Without bootstrap, typical startup voltage is 0.9V.

Construction and testing

An actual-size, single-side PCB layout of the torchlight in normal mode is shown in Fig. 3 and its components layout in Fig. 4. Assemble the circuit on the PCB. Since ZXSC380 is an SMD, place and solder it on the solder side of the PCB. Connect 1.5V cell across CON1.

Similarly, an actual-size PCB layout for the torchlight in bootstrap mode is shown in Fig. 5 and its components layout in Fig. 6. Assemble the circuit on the PCB. Place and solder ZXSC380 on the solder side of the PCB. Connect 1.5V across CON2. EFY

A. Samiuddhin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming